

EW – 1 Lab 2

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VOLTAGE REGULATOR

ZENER DIODE SHUNT VOLTAGE REGULATOR:

A Zener diode is a specially built diode; its depletion layer breaks down (in reverse bias) at low voltages and the characteristic curve post the breakdown voltage shows minimal voltage changes for large current changes. Thus, it acts as a constant voltage regulator over a good range.

For $V_{out} = V_Z$, resistor at input, $R_Z = \frac{V_{in} - V_Z}{I_Z + \frac{V_Z}{L}}$

; $V_Z \rightarrow$ Zener voltage, $I_Z \rightarrow$ Zener current, $L \rightarrow$ Load.

AIM: To make a simple Zener regulator circuit.

Components Required:

- Zener Diode (5.1 V)
- Resistors (100 Ω , 330 Ω , 390 Ω , 1k Ω , 3.3k Ω , 10k Ω)
- Breadboard
- Wires
- Power supply
- Multimeter (DMM)

Procedure:

- Build the voltage regulator by placing the components on the breadboard as per the circuit. Make sure the Zener is reverse biased.
- Provide 15 V input through the power supply and read output into the DMM.
- Choose load resistance as $10\text{k}\Omega$ and change R_z to get different values for I_{in} and V_{out} .
- Next, choose R_z as 330Ω and vary the load. Observe V_{out} as it changes.

Results:

$$R_z = 195.6\Omega \quad (V_{out} = V_z)$$

$$L = 10\text{k}\Omega$$

$R_z (\Omega)$	$V_{out} (V)$	$I_{in} (mA)$
100	5.54	100
330	5.33	30
390	5.30	29.8
1000	5.22	10
3300	5.01	0.036

$$R_z = 330\Omega$$

$L (\Omega)$	$V_{out} (V)$
100	5.29
390	5.30
3300	5.31

$$V_{in} = 15V$$

$$V_{out} = 5.31V$$

$$I_z = 49mA$$

$$\frac{V_z}{L} = \frac{5.31}{10,000} = 0.531mA$$

$$R_z = \frac{V_{in} - V_{out}}{I_z + \frac{V_z}{L}} = \frac{15 - 5.31}{49.531} \times 1000$$

$$= 195.635\Omega$$

Load	$R_z (\Omega)$	$V_{out} (V)$	$I_{in} (mA)$
10k Ω	100	5.54	100
10k Ω	330	5.33	30
10k Ω	390	5.30	29.8
10k Ω	1000	5.22	10
10k Ω	3300	5.01	0.036

$$R_z = \frac{V_{in} - V_z}{I_z + \frac{V_z}{L}} = \text{Remains same}$$

$$(195.6\Omega)$$

$R_z (\Omega)$	Load (Ω)	$V_{out} (V)$
330	100	5.29
330	390	5.30
330	3300	5.31

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Observations:

The output voltage comes close to the Zener voltage when R_z comes closer to the computed value. When the load is varied, the output voltage is more or less maintained, showing that the voltage regulator functions as intended. Errors in the recordings may persist due to calibration errors or loose grips.

Conclusion: A Zener Diode shunt voltage regulator was successfully made.

LM317 ADJUSTABLE POSITIVE SERIES VOLTAGE REGULATOR:

The LM317 package maintains a 1.25V potential difference between its “Out” and “Adjust” pins, and the voltage divider acts as a lever propping V_{out} above the ground reference.

$$V_{out} = (1.25) \cdot \left(1 + \frac{R_2}{R_1}\right)$$

AIM: To make a regulator circuit using LM317 IC.

Components Required:

- LM317 IC
- Resistors (100 Ω , 330 Ω , 390 Ω , 470 Ω , 3.3k Ω , 10k Ω)
- Breadboard
- Wires
- Power supply
- Multimeter (DMM)

Procedure:

- Build the voltage regulator by placing the components on the breadboard as per the circuit. Note the potential between pins 1 and 2.
- Take (15 V) input through the power supply and provide output into the DMM.
- Fix R_1 at 470 Ω and change load resistance to get different values for V_{out} .

Results:

$$R_1 = 470 \, \Omega$$

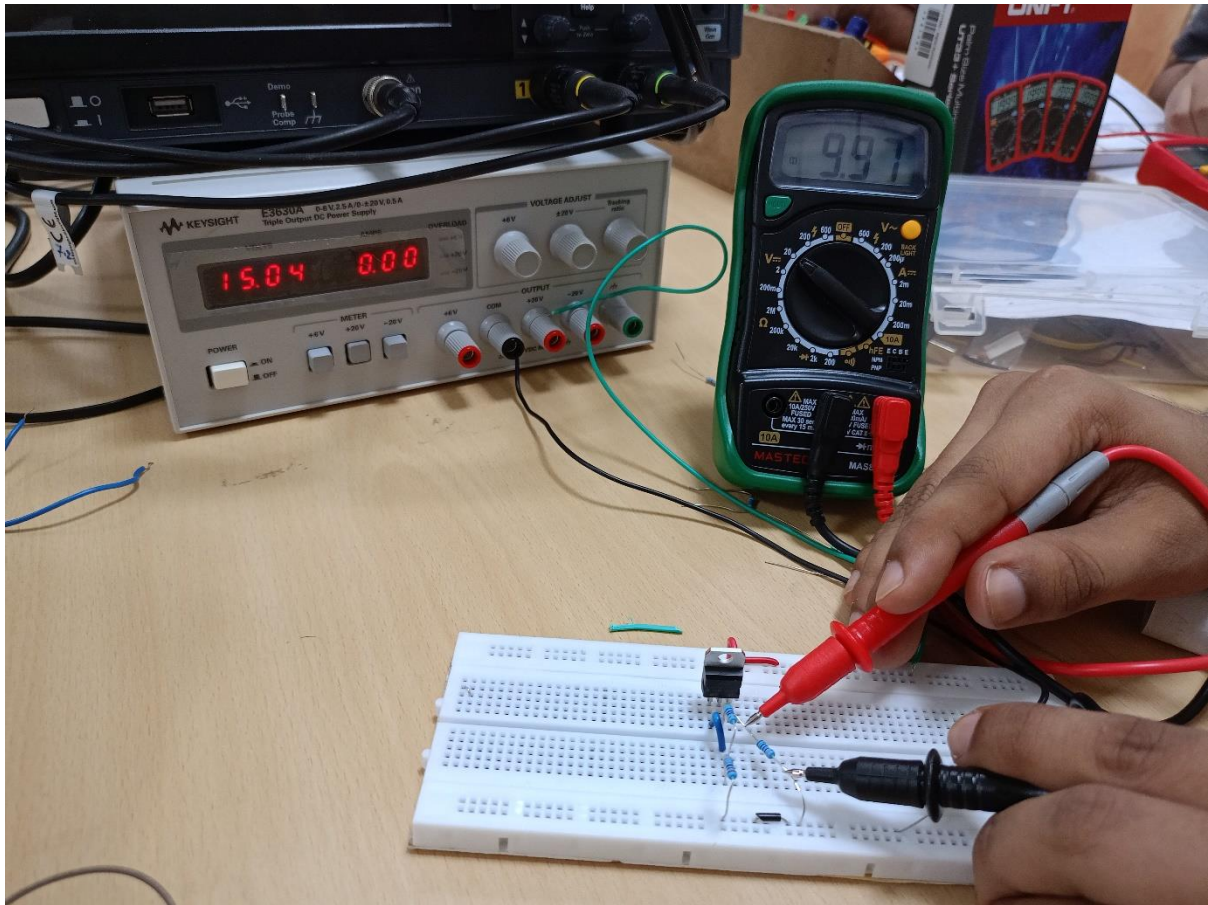
$$R_2 = 3290 \, \Omega$$

$L \, (\Omega)$	$V_{out} \, (V)$
100	9.96
330	9.97
390	9.97
3300	9.96
10000	9.97

① $V_{out} = 1.25 \times \left(1 + \frac{R_2}{R_1}\right)$
 $10 = 1.25 \times \left(1 + \frac{R_2}{R_1}\right)$
 $1 + \frac{R_2}{R_1} = 8$
 $R_2 = 7 \times 470$
 $= 3290 \, \Omega$

$R_1 = 470 \, \Omega$
 $R_2 = 3290 \, \Omega$
Load = $10 \, k\Omega$
(vary)

Load	$V_{out} \, (V)$
100	9.96
330	9.97
390	9.97
3.3k	9.96
10k	9.97



Observations:

V_{out} came out to be close to 10 V, with tiny error due to weak contact of the DMM leads. Change in load resistance did not change the output voltage. R_2 was chosen after fixing R_1 according to the formula prescribed.

Conclusion: A voltage regulator using LM317 was successfully made.

LM7805 VOLTAGE REGULATOR:

The LM7805 package is a constant voltage regulator that outputs 5V at pin 3 for all values between 7V and 35V. Ceramic capacitors are used at the respective pins to reduce ripples in the input/output.

AIM: To test line and load regulation in LM7805 Voltage Regulator IC.

Components Required:

- LM7805 IC
- Resistors (330 Ω , 390 Ω , 470 Ω , 1k Ω , 3.3k Ω , 10k Ω)
- Capacitors (0.1 μ F, 222k pF)
- Breadboard
- Wires
- Power supply
- Multimeter (DMM)

Procedure:

- Place the IC on the breadboard and ground pin 2.
- Ground pin 1 through a 222k pF capacitor, and pin 3 through a 0.1 μ F capacitor.
- Provide input to pin 1 through the power supply. Read pin 3 output via DMM.
- Step up input voltage from 0V to 15V and note V_{out} .
- Next, remove the capacitors and provide input to pin 1 through a 1k Ω resistor. Fix $V_{in} = 15V$.
- Connect a load resistor at pin 3 and vary it to get values for V_{out} .

Results:

$V_{in} (V)$	$V_{out} (V)$
4.5	0.6
6.5	5.24
7.5	5.49
8	5.36
10	5.21
12	5.27
15	4.91

$L (\Omega)$	$V_{out} (V)$
330	4.91
390	4.91
470	4.92
3300	4.92
10000	4.92

~~V_{in}~~
~~4.5~~
~~6.6~~
 7.5
 8.5
 10
 12
 15

~~V_{out}~~

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$$C_1 = 0.1 \mu F$$

$$C_2 = 22 \mu F$$

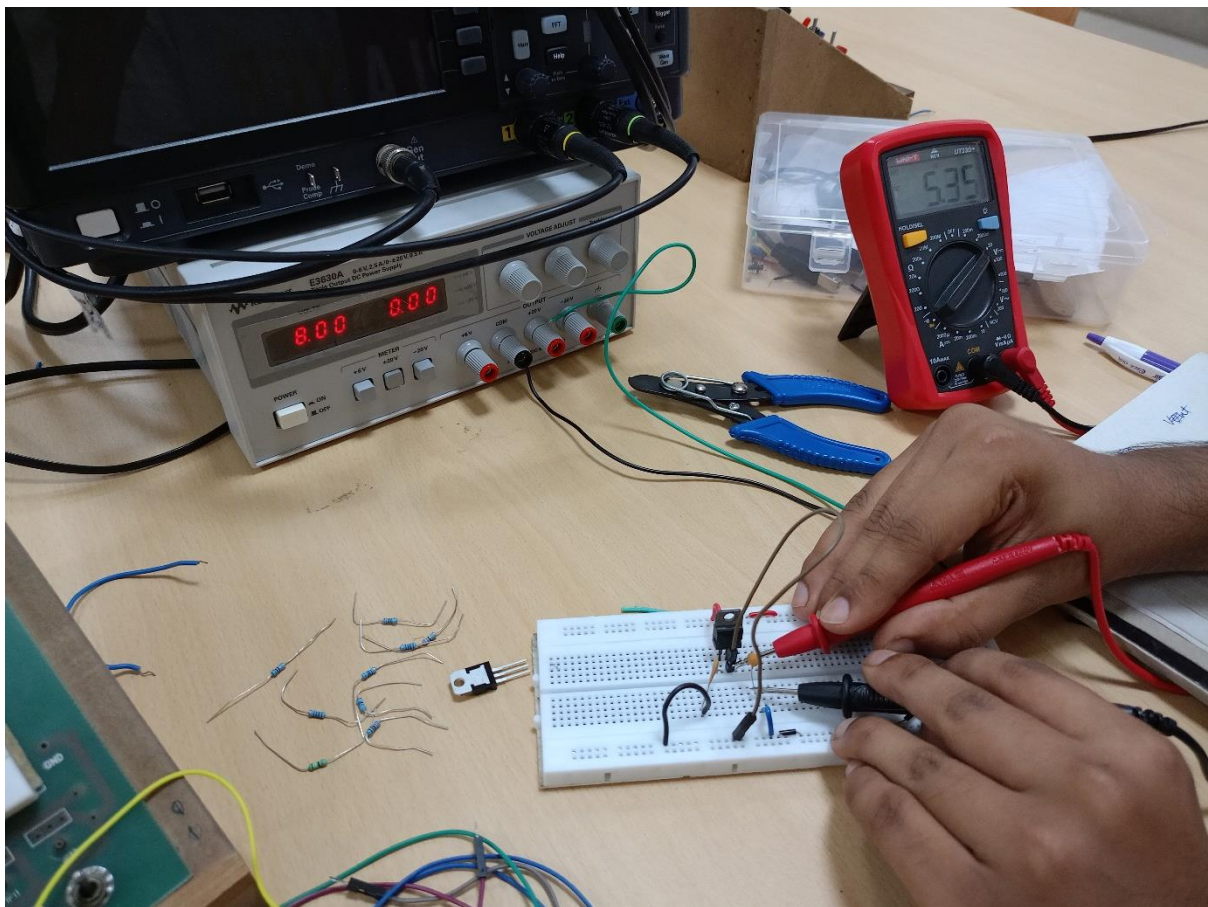
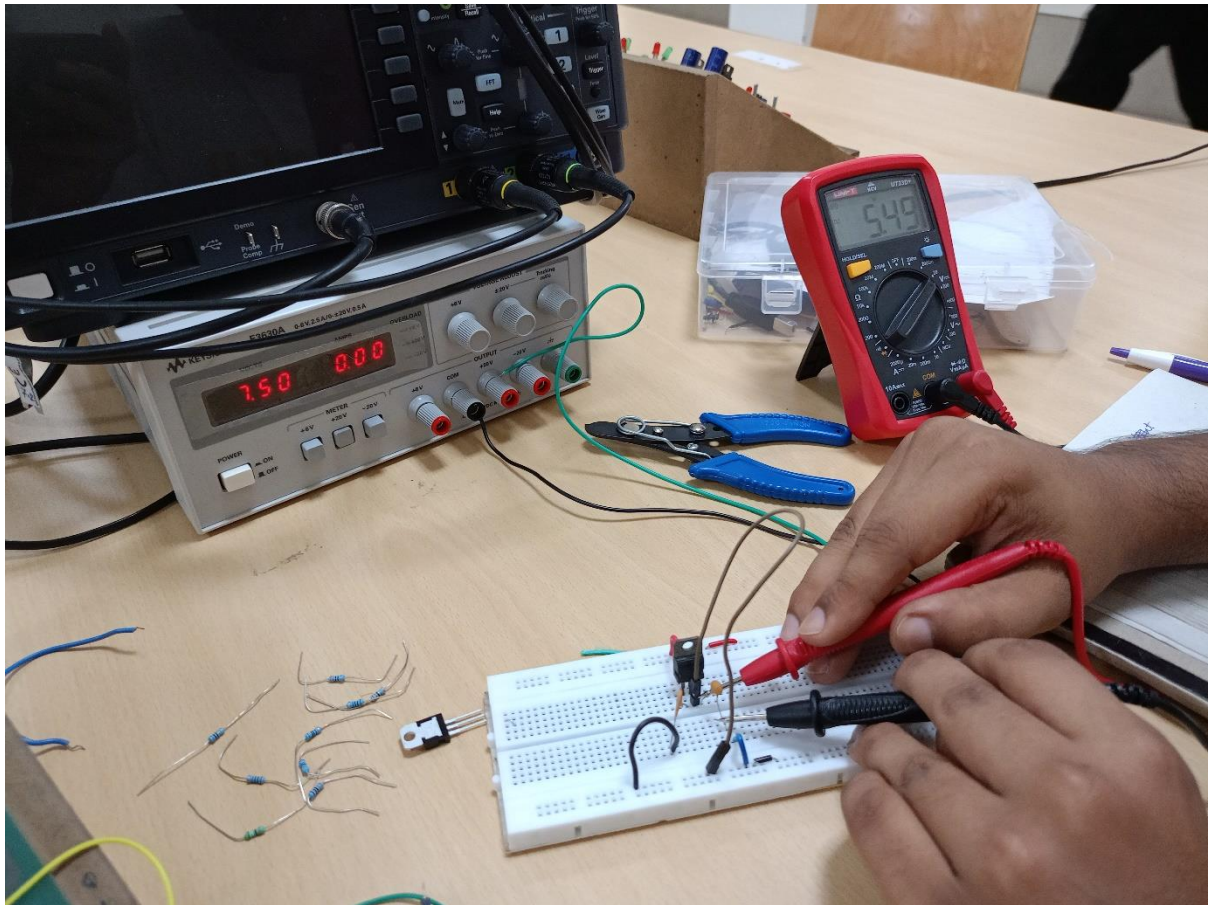
V _{in}	V _{out}
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10	5.21
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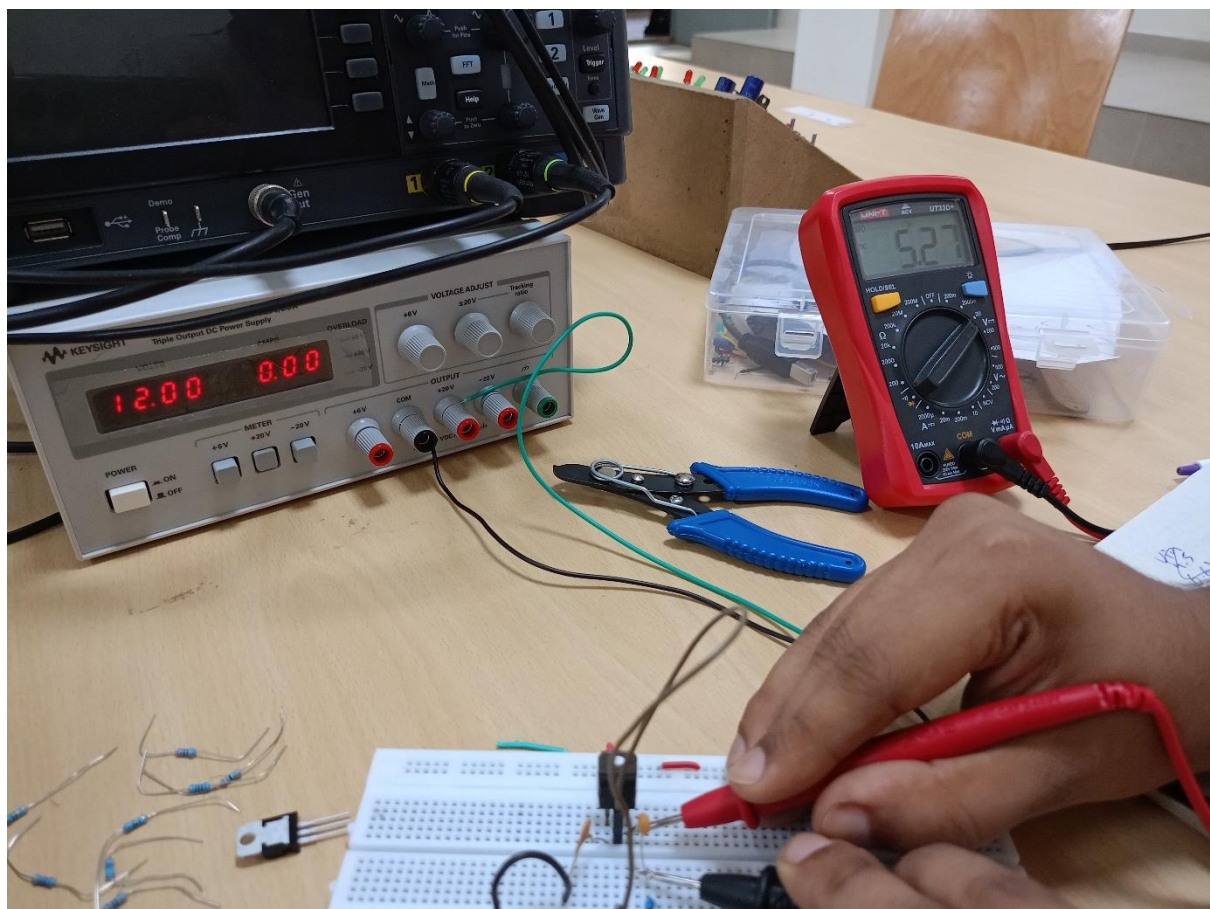
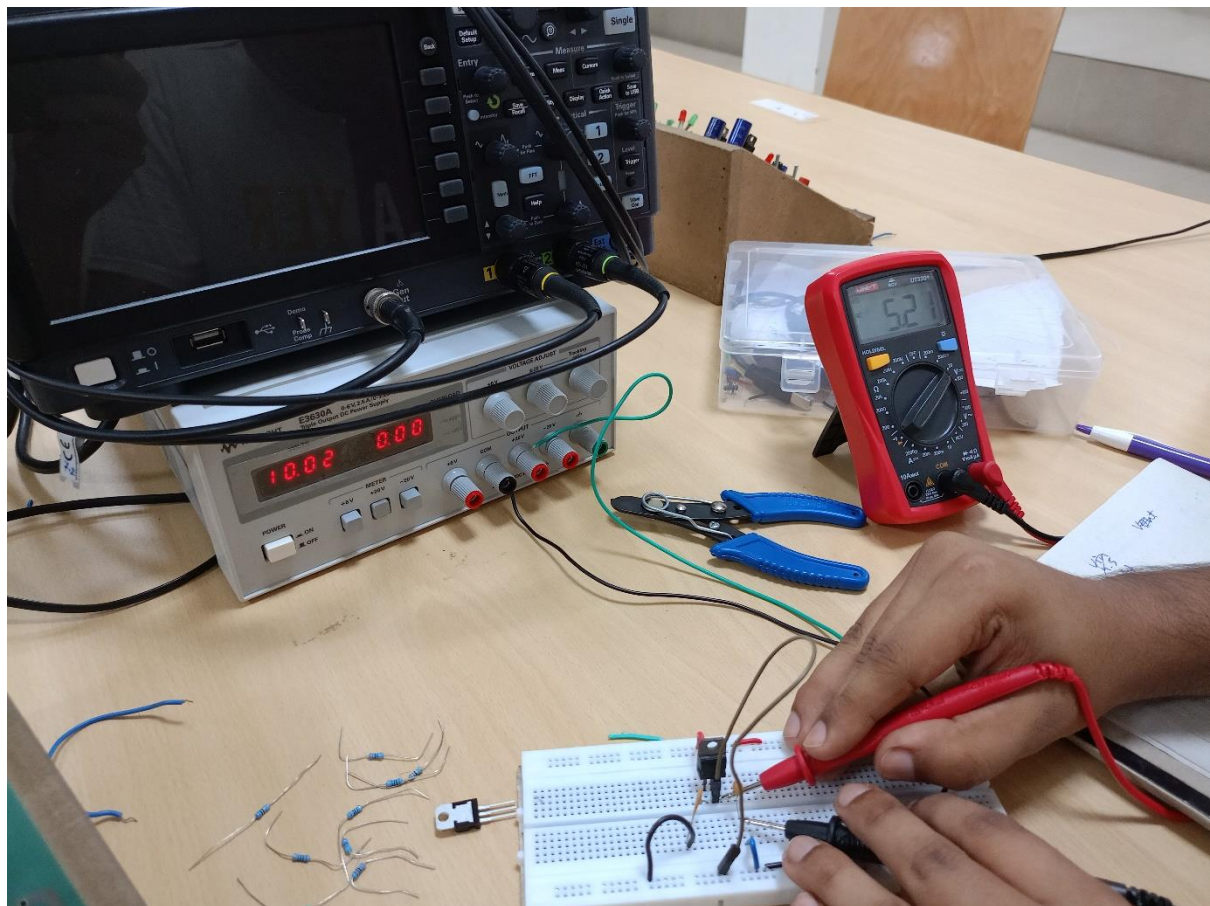
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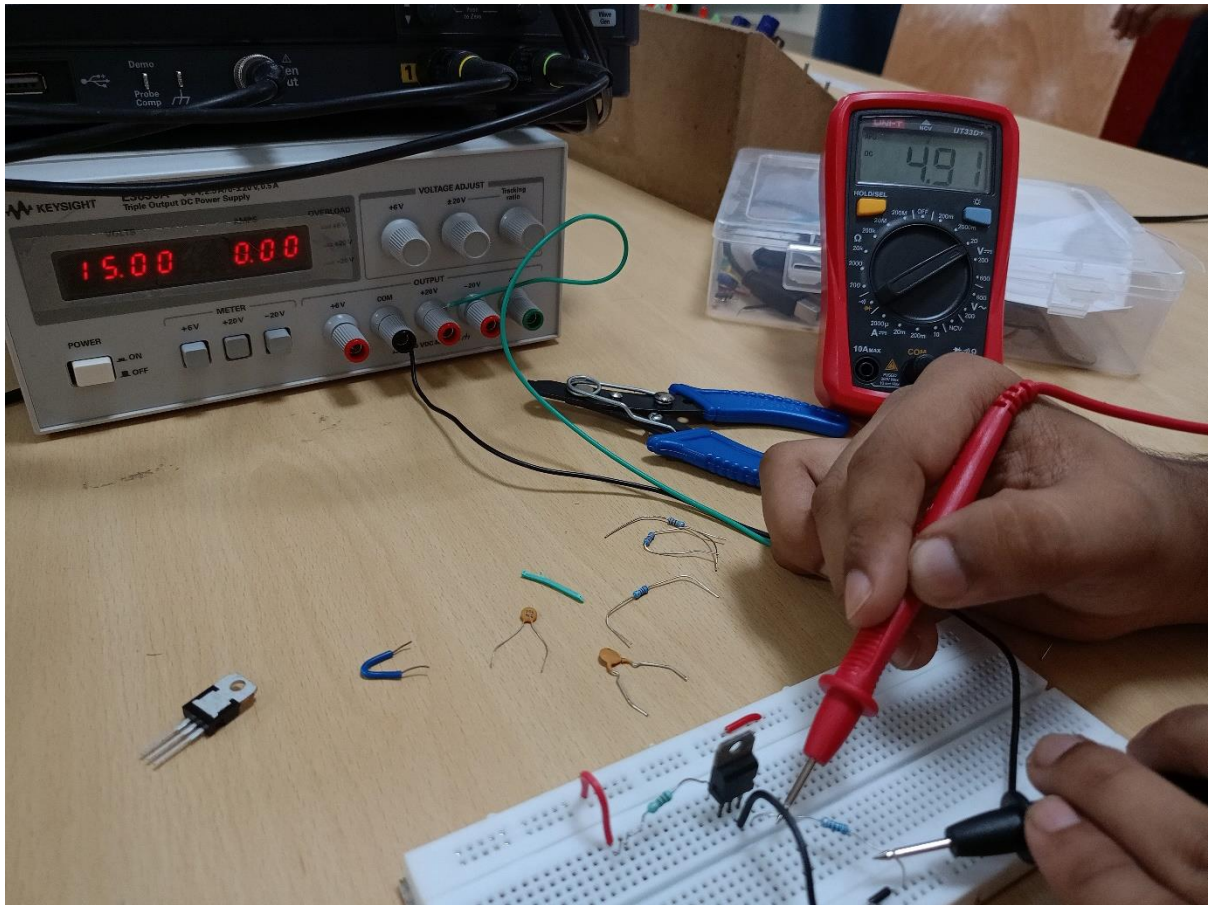
$$R = 1k\Omega \quad V_{in} = 15V$$

Load (Ω)	V _{out}
10k	4.91V
3.3k	4.91V
390	4.92V
330	4.92V 4.89
270	4.92V 4.88

4.89V







Observations:

On varying V_{in} , output voltage changed only marginally, showing line regulation. On changing the load resistance L , the output voltage was more or less constant, i.e., load regulation was successfully established.

Conclusion: The LM7805 voltage regulator package worked as intended.